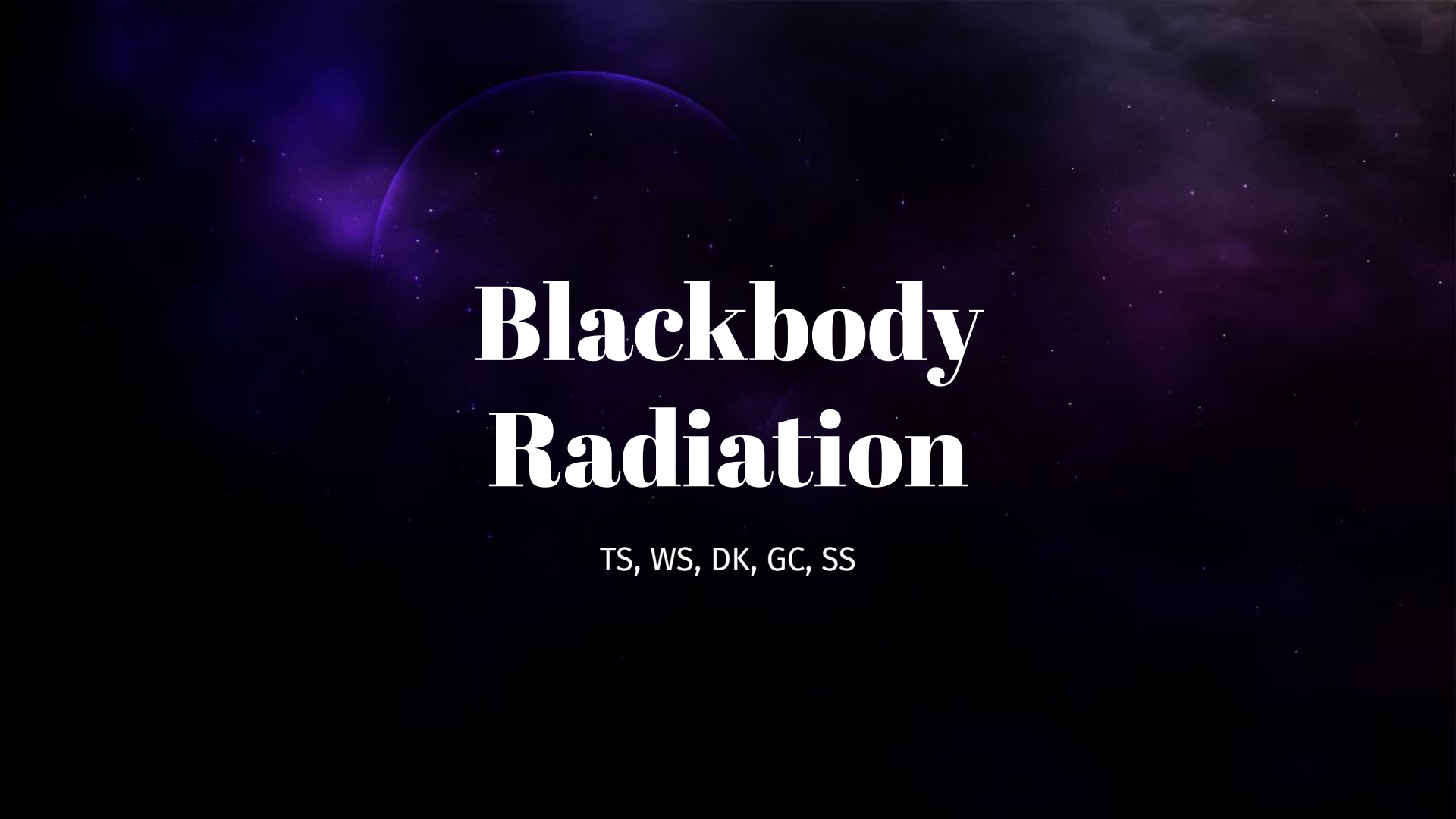


The background of the slide is a deep space scene. It features a large, dark purple planet or moon in the upper left quadrant, partially obscured by a bright, glowing purple nebula. The rest of the background is a dark, starry field with a subtle purple hue. The title text is centered in the middle of the image.

Blackbody Radiation

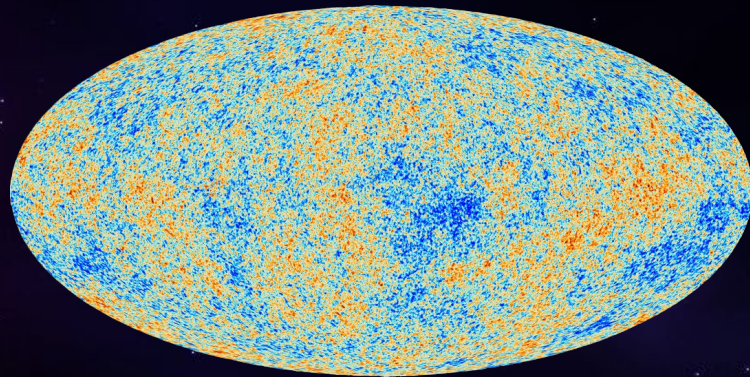
TS, WS, DK, GC, SS



**WHERE DOES
STARLIGHT REALLY
COME FROM?**

What is a Blackbody?

- An Ideal Blackbody is a perfect absorber — a body that absorbs absolutely 100% of light in every spectrum.
- an average “blackbody” today is a star or a black hole
- In reality there is no such thing as a perfect blackbody
- The closest thing that has ever existed to a perfect blackbody is the CMB



What is Blackbody Radiation?

- A perfect absorber is also therefore a perfect emitter
- any form of EM radiation emitted from a blackbody is called blackbody radiation
- This radiation can be most commonly found from stellar objects, such as stars.
- other blackbodies emit less conventional blackbody radiation such as hawking radiation



Why Radiation is Created

Accelerating charged particles emit electromagnetic radiation

Temperature is a measure of average kinetic energy of particles, which means that the particles move around and accelerate (due to collisions caused by temperature)

Simulation from
<https://www.compadre.org/osp/EJSS/4126/154.htm?F=1>



Definition of Variables

c : speed of light

f : frequency

λ : wavelength

u : energy density (per volume)

j : flux (power per unit area)

T : temperature

k : wavenumber

E : energy

B : Radiance (power per unit steradian per unit area) ($j=\pi B$)

Ω : steradian (3D) angles

ϕ : angle of 2D, θ : angle of z-axis

P : pressure, V : volume, U : energy, S : entropy, Ω : number of microstates

k_B : Boltzmann constant

σ : Stefan-Boltzmann constant

b : Wien's constant

h : Planck's constant

On Energy of Light

Photons are isotropic (random direction)

For flux (power per unit area),

- half of the photons face the direction of incident ($\frac{1}{2}$)
- angle of incident is not perpendicular ($\frac{1}{2}$)

This gives flux as a fourth of speed of light * energy density ($j = cu/4$)

For pressure of light,

- radiation arrives at an angle θ upon incident area ($\cos \theta$)
- radiation arrives at an angle θ with momentum ($\cos \theta$)

This gives pressure as a third of energy density ($P = u/3$)

DERIVATION WITHOUT DIRECTION

D

Wien's Displacement Law

$$\lambda_{peak} T = b$$

This equation is derived in order to show that the peak wavelength is inversely proportional to temperature. What that means is that shorter wavelength waves are emitted from hotter objects.

https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_all.html

D

Stefan-Boltzmann Law

$$j = \sigma T^4$$

The Stefan-Boltzmann law tells us that the flux of a blackbody is directly proportional to the fourth power of temperature.

This is the total power emitted per unit area by a blackbody over every wavelength.

Derivation of Stefan-Boltzmann Law (Classical)

From 1st and 2nd laws of thermodynamics:

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P$$

From the facts we know:

$$U = uV$$
$$P = u/3$$

The energy density is proportional to temperature to the fourth power

$$j \propto u \propto T^4$$

Therefore the flux is too ($j = cu/4$). But we do not know the exact value of Stefan-Boltzmann constant from this derivation



**But how does the
graph look like?**

A Universal Curve

Due to energy conservation and second law of thermodynamics,

ALL blackbodies have an universal curve of intensity per wavelength solely depending on their temperature

This does NOT depend on other characteristics of the object
(Kirchhoff's law of thermal radiation)

We already know the peak and area under of the curve (Wien's displacement law, Stefan-Boltzmann law)

DEMONSTRATION ON BOARD

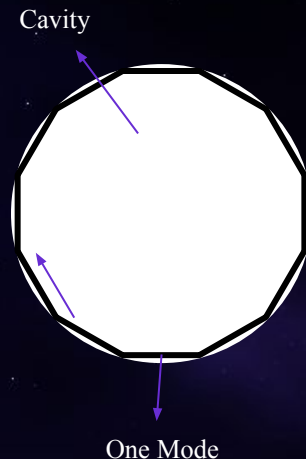
T

How?

energy density at each wavelength = number of possible modes per frequency * average energy density per mode

What are modes?

- Possible wave shape for an electromagnetic wave to have
- Used in blackbodies for standing waves
- Based on wavelength/size of blackbody cavity
- **MUST BE AN EXACT MULTIPLE OF THE WAVELENGTH**

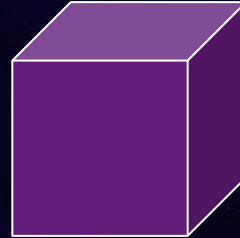


Mode Density

A cubic cavity can be assumed as a blackbody since any light gets absorbed inside

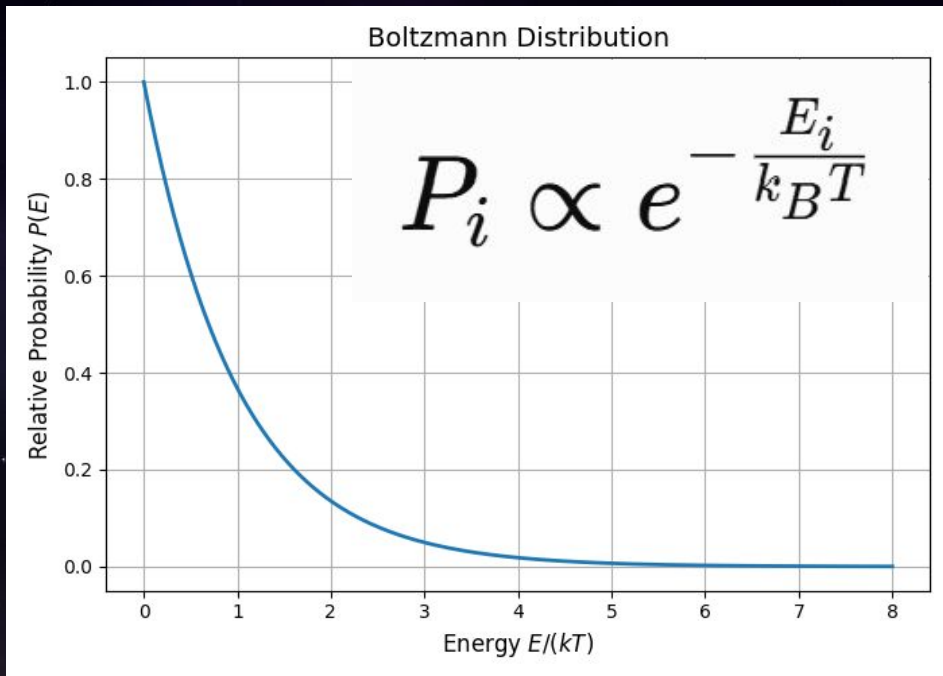
This gives us number of possible modes

$$g(f) = \frac{8\pi f^2}{c^3}$$



What's important is that number of possible modes increase in respect to frequency squared

Boltzmann Distribution

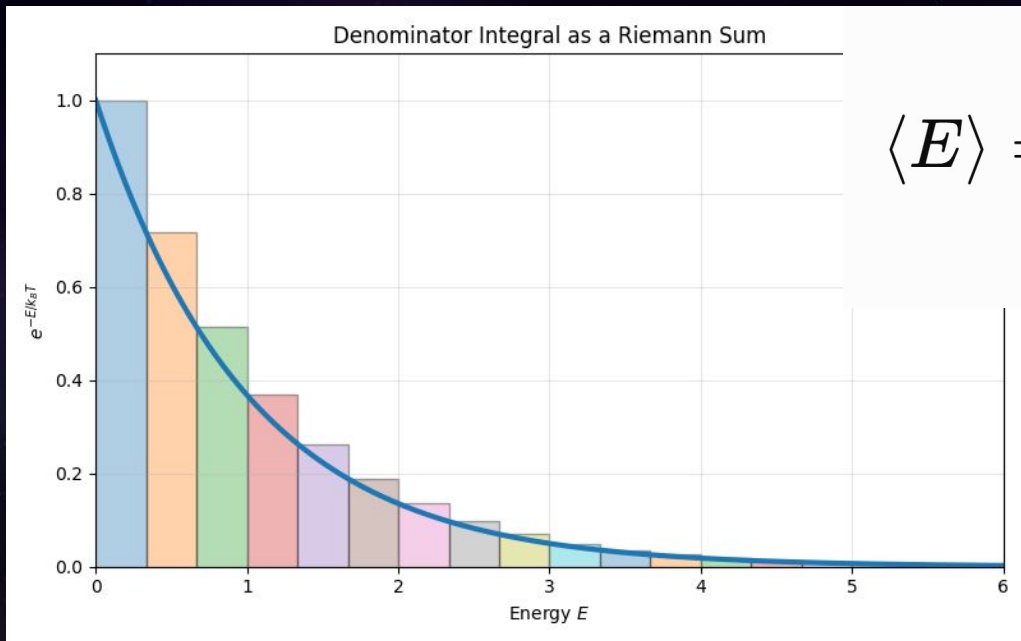


-Boltzmann Distribution:
a probability distribution that determines the likelihood of particles in a system being in a specific energy state.

- Most of particles are in low energy state, few are in high energy state.

- This is due to that there are more possible microstates with low energy

Classical prediction: Average Energy Value



$$\langle E \rangle = \frac{\int_0^\infty E e^{-\frac{E}{k_B T}} dE}{\int_0^\infty e^{-\frac{E}{k_B T}} dE} = k_B T$$

Use integration to calculate average energy per mode, because energy is continuous in classical ways. (weighted average)

Classical prediction: Rayleigh-Jeans Method

energy density per frequency = mode density per frequency * average energy per mode

$$u(f) = \frac{8\pi f^2}{c^3} k_B T$$

By using $j = cu/4$, $df = c/\lambda^2 d\lambda$ (from $f = c/\lambda$)

$$B_\lambda(T) = \frac{2ck_B T}{\lambda^4}$$

This essentially translates to how many possible positions of energy in that area actually have photons travelling through them, as we cannot determine whether each mode has a photon or not easily.

W

Classical prediction: Rayleigh-Jeans Method

$$B_{\lambda}(T) = \frac{2c}{\lambda^4} k \underline{B} T$$

Classical Average Energy



But as wavelength approaches 0, spectral emission approaches infinity

This means that the area under diverges, meaning that every object emitting wavelength under the UV region has infinite power (the integral from 0 to infinity diverges)

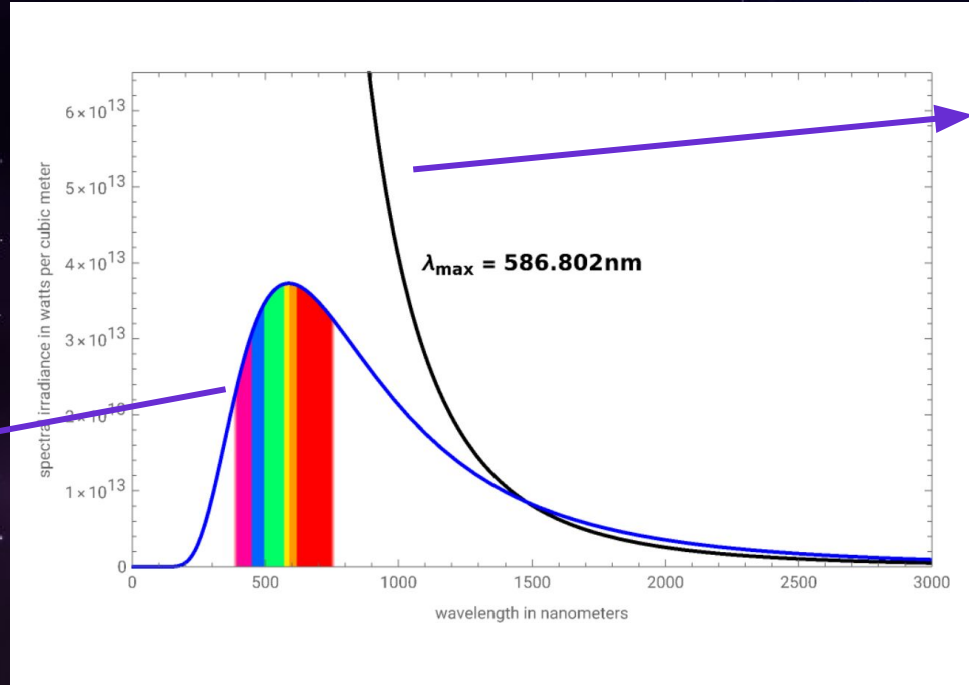
WHICH IS NOT TRUE

W

Classical prediction: Rayleigh-Jeans Method

The Ultraviolet Catastrophe:

The reality



The prediction

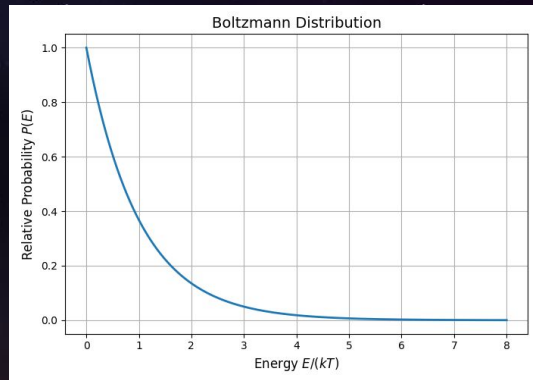
Classical Prediction Logic:

W

Number of modes

$$g(f) = \frac{8\pi f^2}{c^3}$$

Boltzmann Distribution



Average energy per mode

$$\langle E \rangle = \frac{\int_0^\infty E e^{-\frac{E}{k_B T}} dE}{\int_0^\infty e^{-\frac{E}{k_B T}} dE} = k_B T$$

✗

Recognized as wrong and therefore solved by Planck



UV
Catastrophe

Planck's Theory: Quantumization of energy

Based on:

- Boltzmann Distribution
- Modes number calculation

Correction:(Related to energy gap)

- Average energy per mode

Quantization
of energy



E=Total energy of a mode

n=Number of energy levels

h=The Planck Constant

f=The frequency of the photon

$$E = nhf, n = 0, 1, 2, 3, \dots$$

Planck's Theory: Planck Constant

$$h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$$

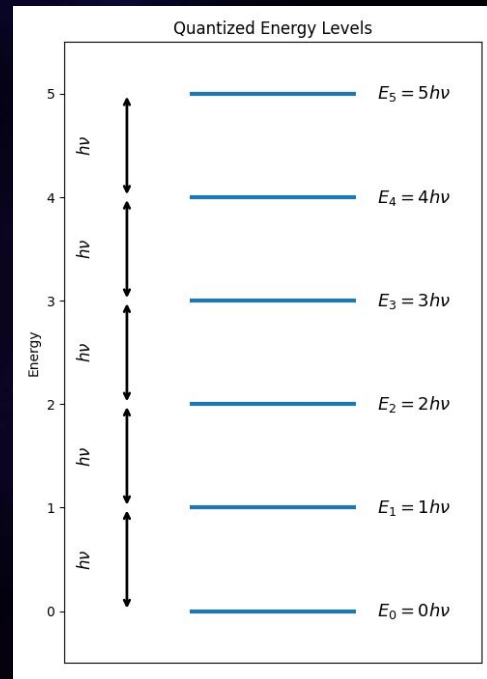
- Comes from experimental fitting
- It sets the fundamental scale of quantum effects.
- Conversion factor between frequency and energy

Planck's Theory: Energy Gap

In classic ways:
there is absolutely no concept of energy gap

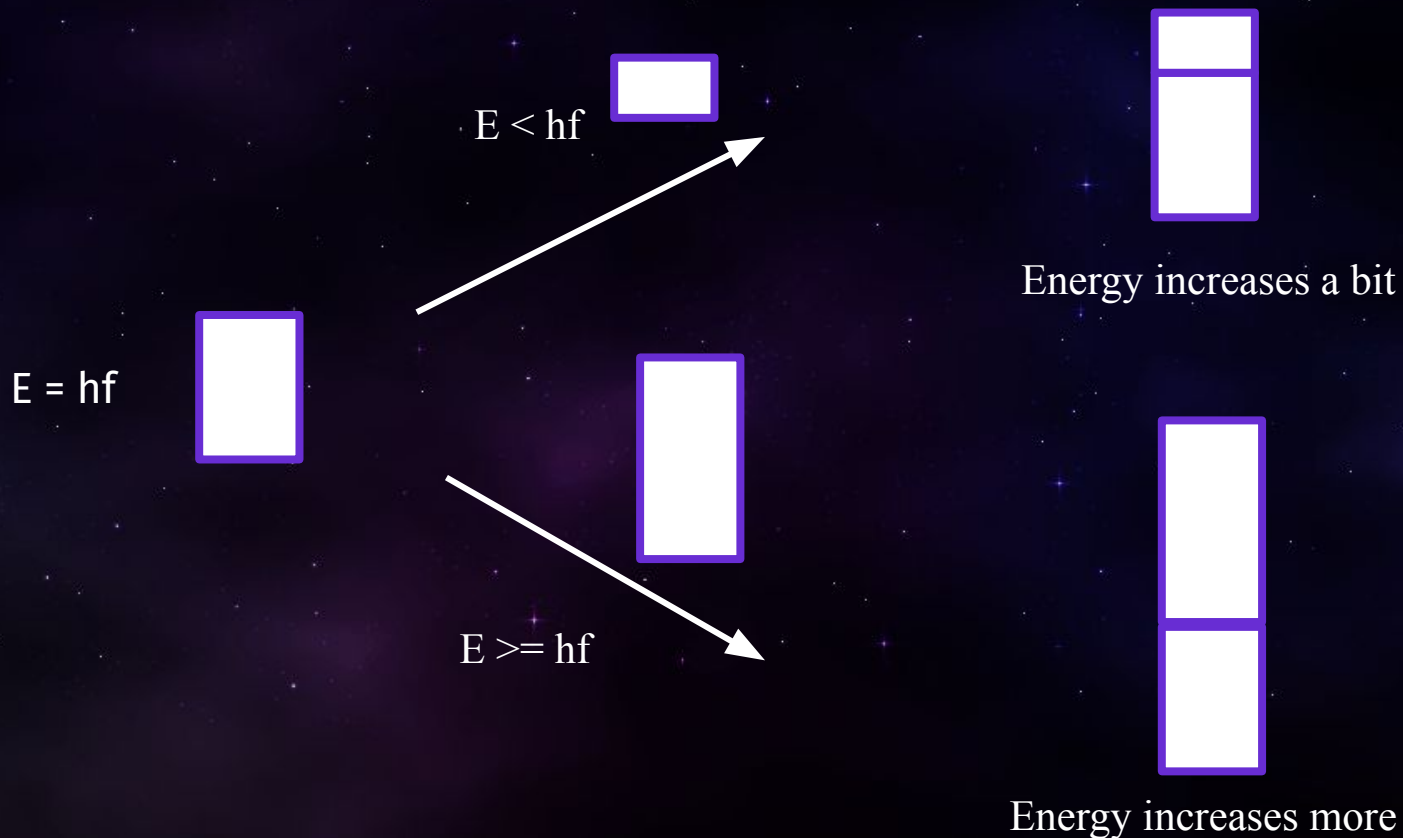
In Planck's theory:
energy of one mode/EM wave increases by steps.
In high frequency area, the energy gap is higher,
so it's hard to be excited.

$$\text{Energy gap} = hf$$



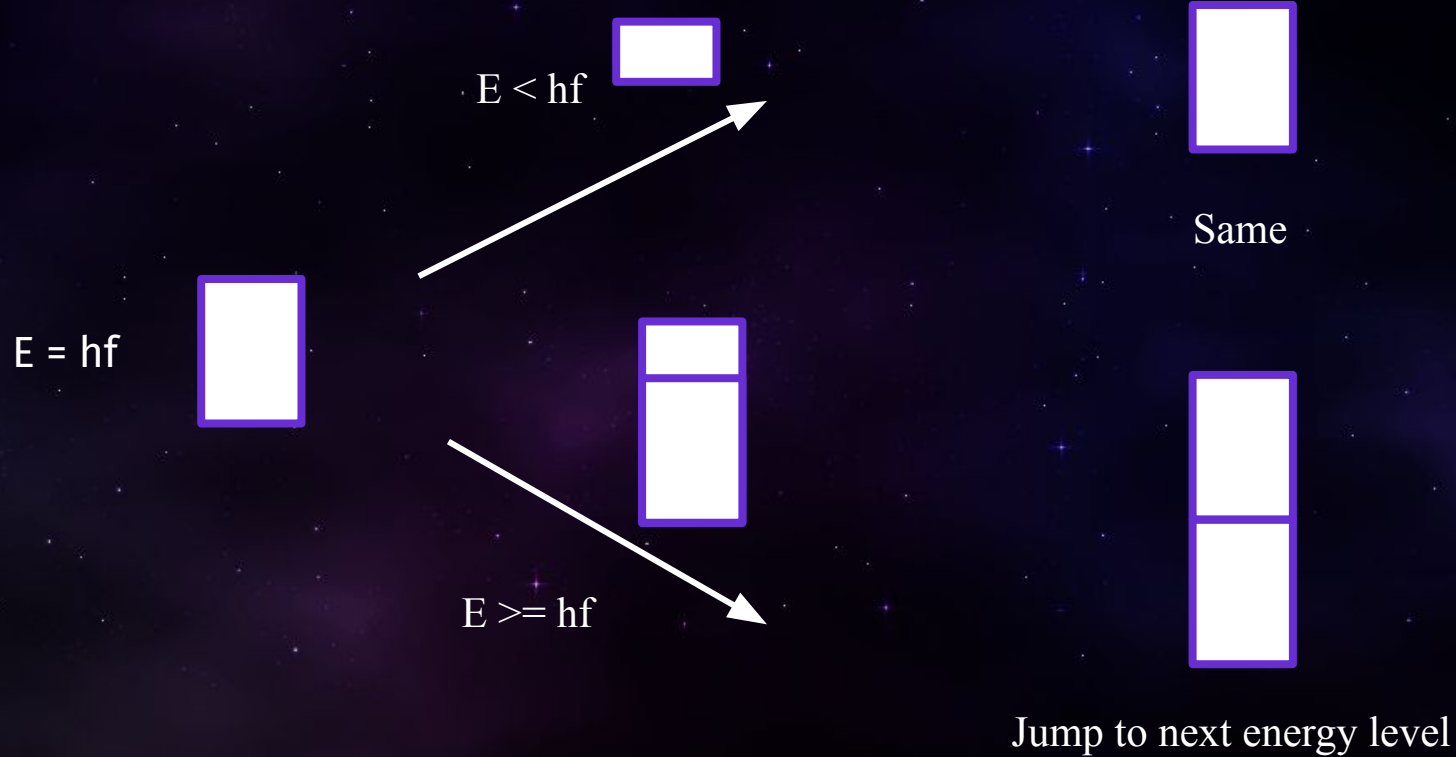
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Classical prediction: Continuous Energy



T

Planck's Theory: Energy Gap



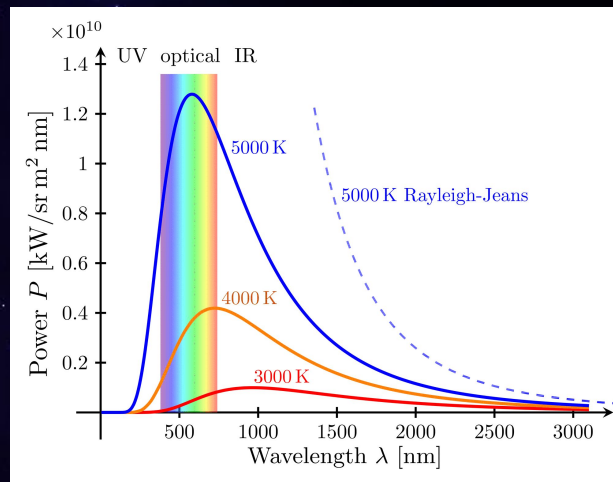
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Planck's Theory: Quantumization of energy

Therefore, we use sums up instead of integration:

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} nhf e^{-nhf/(kT)}}{\sum_{n=0}^{\infty} e^{-nhf/(kT)}} = \frac{hf}{e^{hf/(kT)} - 1}$$

$$B_{\lambda}(T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda kT)} - 1}$$



W

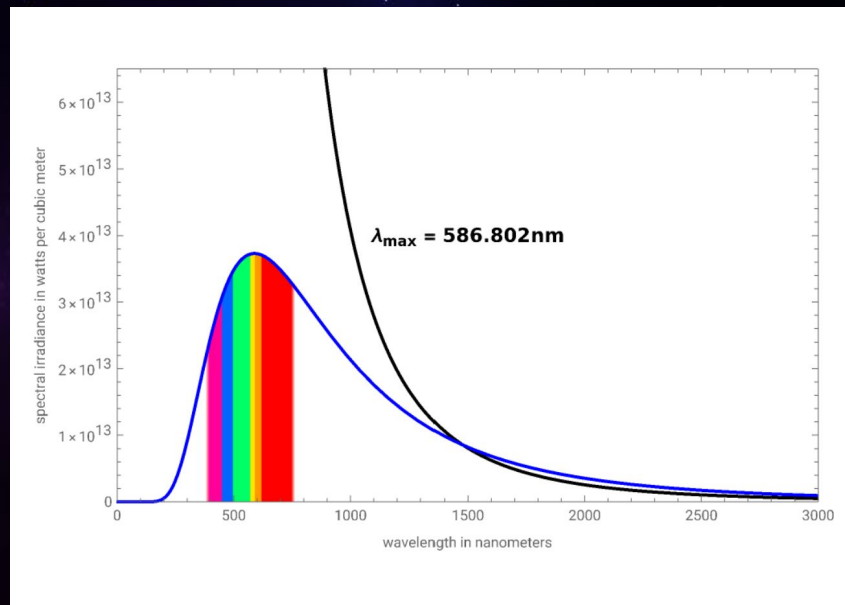
Connection between Classics and Planck

$$e^{\frac{hc}{\lambda k_B T}} \approx 1 + \frac{hc}{\lambda k_B T}$$

$$B(\lambda, T) \approx \frac{2hc^2}{\lambda^5} \cdot \frac{1}{\frac{hc}{\lambda k_B T}}$$
$$= \frac{2ck_B T}{\lambda^4}$$



Rayleigh-Jeans Law in
low frequency area



D

Derivation of Stefan-Boltzmann Law (Quantum)

Get the total area under the curve of Planck's equation (integrate from 0 to infinity)

The result gives total flux of

$$j = \int_0^{\infty} \left(\frac{2hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1} \right) d\lambda = \frac{2\pi^5 k_B^4}{15h^3 c^2} T^4 = \sigma T^4$$

DIMENSIONAL ANALYSIS

D

Derivation of Wien's Displacement Law

If we get the maximum intensity with Planck's equation

we are able to know the product of peak wavelength and temperature is always constant

$$B_{\lambda}(T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1}$$

$$y = \frac{hc}{\lambda k_B T}$$

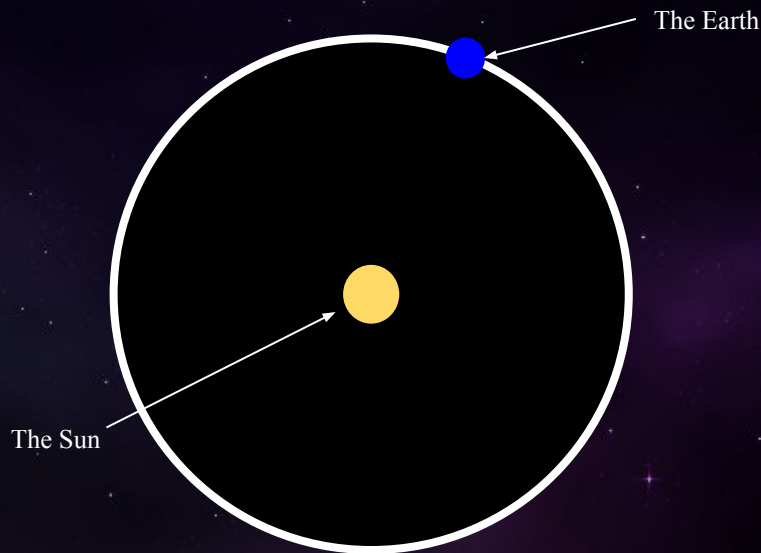
$$B \propto \frac{y^5}{e^y - 1}$$

$$\frac{d}{dy} \left(\frac{y^5}{e^y - 1} \right) = 0 \Rightarrow y = 5(1 - e^{-y}) \Rightarrow y \approx 4.965$$

$$\lambda_{max} T = \frac{hc}{y k_B} = b \approx 2.898 \times 10^{-3}$$

DIMENSIONAL ANALYSIS

Estimation: Temperature of Earth



Power emitted by the Sun: $L_s = 4\pi R^2 \cdot \sigma T^4$

Solar flux on the Earth: $F = \frac{L_s}{4\pi d^2}$

Because the Earth is not a perfect black body, it will reflect some portion of the light.

A reflection coefficient α is introduced.

We introduce a Power absorbed by the Earth:

$$P_s = F \cdot \pi R_E^2 \cdot (1 - \alpha) = \frac{L_s R_E^2 (1 - \alpha)}{4d^2}$$

Estimation: Temperature of Earth

Power emitted by the Earth: $P_e = 4\pi R_E^2 \cdot \sigma T_E^4$

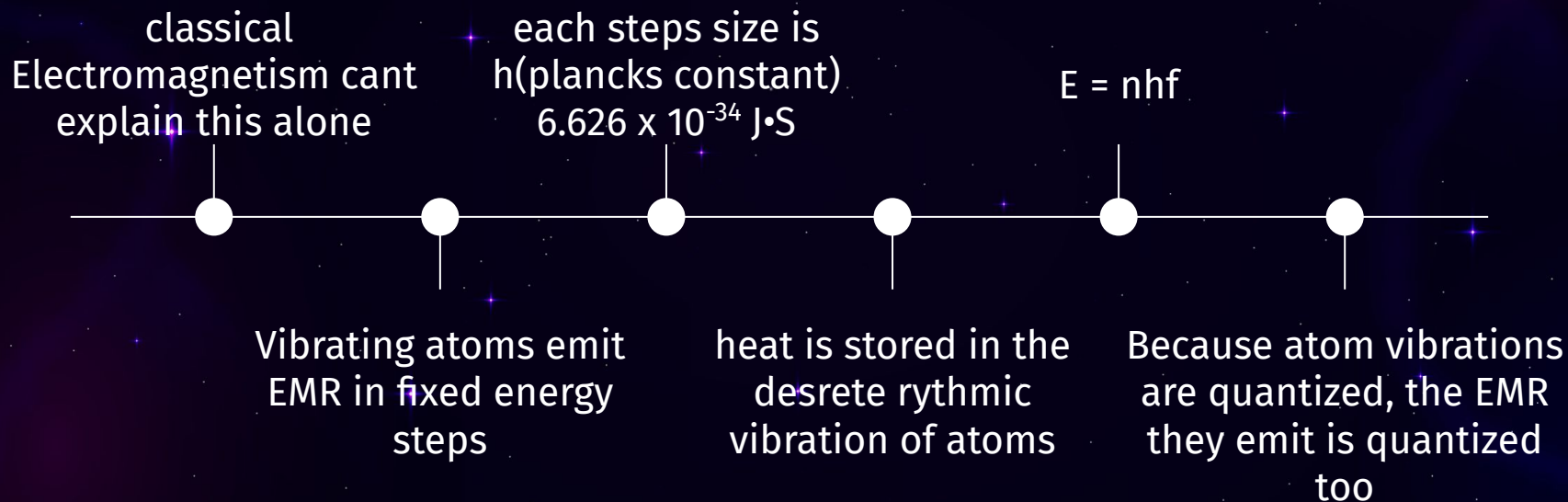
The Earth is in thermal equilibrium: $P_a = P_e$

Plug in numbers, we get $T_E \approx 255K$, which is about -18°C .

It's apparently not the normal temperature of the Earth.

That is to say, greenhouse gas shields the earth and reserves some heat.

Planck's Constant



Appendices

- Energy of light
- Mode density
- Boltzmann Thermodynamics Derivation
- Boltzmann Distribution Derivation

Energy of Light Appendix

$$j = \frac{\text{Power}}{A} = \frac{u^* A^* l}{A^* t} = uc$$

$$du = \frac{u}{4\pi} d\Omega$$

$$d\Omega = \sin\theta d\theta d\phi$$

$$dj = \frac{cu}{4\pi} \cos\theta d\Omega$$

$$0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \phi \leq 2\pi$$

$$j = \frac{cu}{4\pi} \int_0^{2\pi} d\phi \int_0^{\pi/2} \cos\theta \sin\theta d\theta = \frac{cu}{4\pi} (2\pi)(1/2) = \frac{cu}{4}$$

Energy of Light Appendix

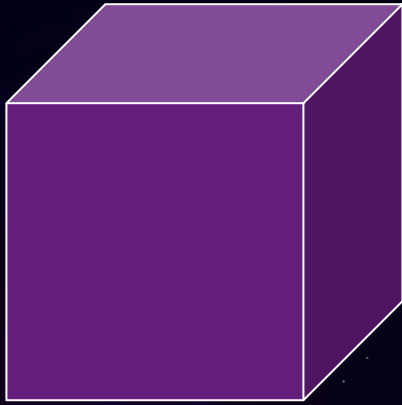
$$E^2 = (mc^2)^2 + (pc)^2 \rightarrow E = pc$$

$$Pressure = \frac{F}{A} = \frac{p/t}{A} = \frac{p}{At} = \frac{E}{Atc} = u$$

$$P = \overline{ucos^2\theta}, \text{ weighted by } sin\theta d\theta \text{ due to sphere shape}$$

$$\overline{cos^2\theta} = \frac{\int_0^{\pi/2} cos^2\theta sin\theta d\theta}{\int_0^{\pi/2} sin\theta d\theta} = 1/3$$

Mode Density Appendix



L

The radiation in the cavity is a collection of standing electromagnetic waves. The electric field should be perpendicular at the surface of conductors. In a cavity of conducting cube (length L)

$$1D: L = \frac{n\lambda}{2}; 3D: L = \frac{n_x\lambda}{2}, \frac{n_y\lambda}{2}, \frac{n_z\lambda}{2}, (n = 1, 2, 3, \dots)$$

$$k_x = \frac{n_x\pi}{L}, k_y = \frac{n_y\pi}{L}, k_z = \frac{n_z\pi}{L}, k_{vector} = k_x + k_y + k_z$$

In a k-space, points exist every π/L for all axis in the positive region

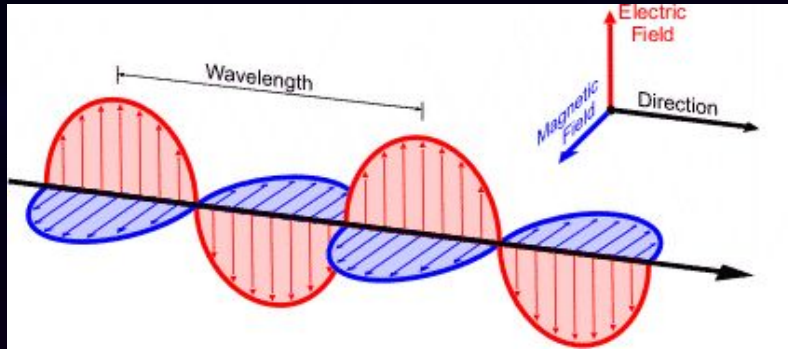
This makes a point take up $(\pi/L)^3$ volume. So the density of point is V/π^3

And the number of these points in a small dk with a sphere only in the positive region,

$$k + dk = \frac{1}{8} * 4\pi k^2 dk = \frac{1}{2}\pi k^2 dk \text{ (volume)}$$

$$dN = (V/\pi^3) * \frac{1}{2}\pi k^2 dk = \frac{Vk^2}{2\pi^2} dk$$

Mode Density Appendix



But as electric waves have two possible directions with a single direction of the light,

$$dN = \frac{vk^2}{\pi^2} dk$$

$$c = f\lambda, k = 2\pi/\lambda = 2\pi f/c, dk = 2\pi/c df$$

$$dN = \frac{8\pi f^2 V}{c^3} df$$

The density of possible modes with frequency $f + df$ comes out

$$g(f)df = \frac{8\pi f^2}{c^3} df$$

Boltzmann Thermodynamics

$$\text{1st: } Q = dU + PdV, \text{ 2nd: } TdS = Q$$

$$\Rightarrow dU = TdS - PdV$$

$$U(T, V), dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV; S(T, V), dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV$$

$$dU = T \left[\left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV \right] - PdV$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T \left(\frac{\partial S}{\partial V}\right)_T - P$$

$$\text{Maxwell relation: } \left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T \left(\frac{\partial P}{\partial T}\right)_V - P$$

Boltzmann Distribution Derivation

Bath

Mode

$$E_{total} = E_{bath} + E_{mode} \quad (E_{mode} = E)$$

The number of possible microstates decrease for bath as E increases since the Energy of bath decreases and there is less ways to distribute smaller energy among same number of multiple modes in bath

And “Every microscopic state of the combined system is equally likely.”

$$\Rightarrow P(E) \propto \Omega_{bath}(E_{bath})$$

$$S = k_B \ln(\Omega), \quad \Omega = e^{S/k_B}$$

$$S(E_{bath}) = S(E_{total} - E) \approx S(E_{total}) - E \left(\frac{\partial S}{\partial E} \right), \quad \frac{\partial S}{\partial E} = 1/T$$

$$S(E_{bath}) = S(E_{total}) - E/T$$

$$P(E) \propto e^{-E/k_B T}$$